

Toxicity Report – Gliss Ultimate Repair Shampoo Ingredients

This report summarises publicly available safety and toxicity information for each listed ingredient in this shampoo. It is not a diagnosis or personalised medical advice. For individual concerns (allergies, pregnancy, existing scalp disease, etc.), consult a doctor or dermatologist.

1. Summary Table

Ingredient	Main Role	Known Toxicity / Concerns at Cosmetic Levels	Practical Risk (Normal Adult Scalp/Hair Use)
Aqua (Water)	Solvent	No relevant toxicity concerns.	Very low
Sodium Laureth Sulfate	Primary anionic surfactant/cleanser	CIR and other reviews: safe as used in rinse-off products, but a known skin and eye irritant at higher concentrations; not a sensitiser.	Low-moderate (dryness/irritation if scalp is sensitive).
Cocamidopropyl Betaine	Amphoteric surfactant, foam booster	Safety panels: considered safe in current use; can cause mild irritation and, in some people, allergic contact dermatitis related partly to impurities.	Low-moderate (irritation/allergy in a minority).
Sodium Chloride	Viscosity adjuster (thickener)	Inert salt at cosmetic levels; high levels may be drying or sting broken skin.	Very low-low
Dimethicone	Silicone conditioning agent	CIR: dimethicone/methicone polymers are safe as used in cosmetics with very low systemic toxicity and weak irritation.	Very low
Hydrolyzed Keratin	Protein-derived hair conditioning agent	Large peptides with minimal penetration; main concern is rare allergy.	Very low-low
Hydrolyzed Pearl	Mineral/protein complex, shine and marketing ingredient	Very limited systemic absorption; toxicity concerns are minimal; allergy is possible but rare.	Very low-low
PEG-7 Glyceryl	Mild non-ionic	PEGylated oils generally	Low

Cocoate	surfactant/emollient	considered safe; available data for this specific ingredient show low acute toxicity, though human data are limited.	
Parfum (Fragrance)	Fragrance blend	Fragrances are a leading cause of cosmetic allergy and irritation; composition not fully disclosed.	Moderate for fragrance-sensitive users
Sodium Benzoate	Preservative	Widely used in foods/cosmetics; low systemic toxicity but can irritate very sensitive skin or mucosa.	Low
Amodimethicone	Modified silicone for conditioning and smoothing	CIR: amodimethicone and related silicones considered safe as used in cosmetics with low toxicity.	Very low
Citric Acid	pH adjuster, mild AHA	At typical shampoo levels mainly controls pH; higher strengths can irritate and increase photosensitivity.	Low
Guar Hydroxypropyltrimonium Chloride	Cationic polymer conditioner	CIR and other reviews: non-sensitising, low toxicity, considered safe as used.	Very low
Hydrogenated Castor Oil	Emulsifier / stabiliser	Low toxicity; rare cases of allergy reported.	Very low–low
Glyceryl Distearate	Emulsifier, opacifier	Low dermal absorption; generally regarded as safe.	Very low
Laureth-4	Non-ionic surfactant, solubiliser	Ethoxylated surfactant; irritation potential at high levels; theoretical concern about 1,4-dioxane impurities if poorly purified.	Low–moderate (mainly irritation).
PEG-120 Methyl Glucose Dioleate	Thickener/surfactant	CIR safety report: very low acute toxicity (oral LD50 > 5 g/kg) and considered safe as used in cosmetics.	Low
Laureth-23	Non-ionic surfactant, solubiliser	Similar profile to Laureth-4; low systemic toxicity; potential for irritation at higher concentrations.	Low–moderate
Trideceth-10	Non-ionic surfactant	Used at low levels; irritation	Low

		possible in sensitive individuals; systemic toxicity low.	
Linalool	Fragrance component (floral)	Recognised EU fragrance allergen; oxidised forms particularly sensitising.	Moderate for perfume-allergic users
Sodium Hydroxide	pH adjuster (alkali)	Highly caustic when concentrated; at the tiny amounts used only to adjust pH, systemic toxicity is negligible, but it can sting if product enters eyes.	Low
Limonene	Fragrance component (citrus)	Recognised fragrance allergen; oxidised limonene is a strong sensitiser.	Moderate for perfume-allergic users
Phenoxyethanol	Preservative	EU SCCS: safe up to 1% in cosmetics with a good safety margin; main issues are irritation or allergic dermatitis in a minority.	Low-moderate
Propylene Glycol	Humectant, solvent	Generally recognised as safe at cosmetic levels; can cause irritation or allergy in a small percentage of people.	Low
Sodium Salicylate	Preservative / anti-inflammatory agent	At cosmetic levels systemic exposure is very low; can irritate very sensitive skin and is related to salicylate allergy/aspirin sensitivity.	Low-moderate (if salicylate-sensitive).
Potassium Sorbate	Preservative	Good overall safety profile; occasional local irritation/allergy.	Low
Sorbic Acid	Preservative (acid form corresponding to potassium sorbate)	Similar to potassium sorbate; low systemic toxicity, occasional irritation.	Low
Benzyl Alcohol	Fragrance component and preservative booster	Recognised fragrance allergen; can cause irritation or allergic dermatitis in susceptible individuals.	Moderate for fragrance/allergy-prone users

2. Detailed Notes by Ingredient Group

A. Cleansers and Surfactants

Sodium laureth sulfate (SLES), cocamidopropyl betaine, laureth-4, laureth-23, PEG-7 glyceryl cocoate and trideceth-10 make up the main cleansing and foaming system. SLES is a very common anionic surfactant. The Cosmetic Ingredient Review (CIR) Expert Panel has concluded that sodium laureth sulfate is safe as used in cosmetics, although it is a recognised skin and eye irritant, especially at higher concentrations or with prolonged contact. In rinse-off shampoos, irritation risk is reduced but still relevant for sensitive scalps.

Cocamidopropyl betaine (CAPB) is an amphoteric surfactant used to boost foam and reduce the harshness of SLES. Both CIR and more recent toxicology reviews have found CAPB to be safe at cosmetic use levels, but patch-test data show it can cause mild irritation and, in some individuals, allergic contact dermatitis, often linked to residual impurities from its manufacture. The non-ionic surfactants laureth-4, laureth-23, trideceth-10 and PEG-7 glyceryl cocoate are generally low in systemic toxicity but can contribute to irritation at high levels; like other ethoxylated ingredients they may contain trace amounts of 1,4-dioxane if purification is poor.

B. Conditioning Silicones and Polymers

Dimethicone and amodimethicone are silicone polymers that coat the hair shaft to improve smoothness and shine. The CIR Expert Panel has reviewed dimethicone, amodimethicone and related methicones and concluded they are safe as used in cosmetics, with very low dermal penetration, low systemic toxicity and minimal irritation potential. Guar hydroxypropyltrimonium chloride is a cationic polymer derived from guar gum that improves slip and reduces static. Published CIR findings and industry safety summaries describe it as non-sensitising and of low toxicity at the low concentrations used in hair products.

Hydrolyzed keratin and hydrolyzed pearl are large protein or mineral complexes used mainly for marketing and minor conditioning effects; they have minimal skin penetration and low systemic toxicity, with allergy as the main theoretical concern. Hydrogenated castor oil and glyceryl distearate act as emulsifiers and stabilisers and are also regarded as low-toxicity, low-irritation ingredients.

C. pH Adjusters, Solvents and Humectants

Citric acid and sodium hydroxide are used together to fine-tune the shampoo's pH. At the tiny concentrations needed for pH adjustment, systemic toxicity is negligible, but both substances can sting if product enters the eyes. Propylene glycol and glycerin (if present) act as humectants and solvents; both have good safety records in cosmetics, although a small percentage of people experience irritation or allergic contact dermatitis from propylene glycol. Sodium chloride adjusts viscosity and can be mildly drying at higher levels.

D. Preservatives

Sodium benzoate, phenoxyethanol, sodium salicylate, potassium sorbate and sorbic acid form the preservation system. Regulatory bodies such as the EU Scientific Committee on Consumer Safety consider phenoxyethanol safe at up to 1% in cosmetic products, with a comfortable safety margin based on dermal exposure. Sodium benzoate, sorbic acid and potassium sorbate have long histories of safe use in food and cosmetics. Sodium salicylate is related to salicylic acid; at the low levels used for preservation, systemic exposure is very small, though people with known salicylate/aspirin sensitivity should be cautious. Across all of these, the practical concerns are local irritation or allergic reactions in susceptible users rather than serious systemic effects.

E. Fragrance and Fragrance Allergens

The generic 'parfum' denotes a proprietary fragrance blend whose exact composition is not disclosed. In addition, linalool, limonene and benzyl alcohol are listed separately. These substances belong to the group of 26 fragrance allergens identified by the European Union as having a relatively high rate of allergic contact dermatitis. By law, they must be declared on product labels when present above very low thresholds. At cosmetic levels they are not systemically toxic, but they are a significant cause of perfume-related rashes and itching. For individuals with fragrance allergy, this fragrance system represents the highest practical risk in the formula.

F. Overall Assessment

For a typical adult user with a healthy scalp, this shampoo's ingredient list is consistent with modern commercial formulations: surfactants, conditioning silicones and polymers, standard preservatives and a scented fragrance blend. None of the ingredients are recognised human carcinogens at cosmetic concentrations, and expert safety panels have reviewed key components such as sodium laureth sulfate, cocamidopropyl betaine, dimethicone, amodimethicone, PEG-120 methyl glucose dioleate and guar hydroxypropyltrimonium chloride and found them safe as used in rinse-off products. The real-world risks are local: dryness or irritation from the cleansing system, and fragrance-related allergy. Users with a history of eczema, very sensitive scalp or perfume allergy may wish to limit exposure or choose a fragrance-free, milder-surfactant shampoo.